6.5-function-visualisation

June 10, 2026

1 Function Visualisation

[ ]:
```python
import polars as pl
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

PARQUET_PATH = "../../data/interim/encoded-collapsed-waveform.parquet"

df = pl.read_parquet(PARQUET_PATH)

nozzle_cols = [c for c in df.columns if c.startswith("Value_") and c.
 ↪endswith("#_mean")]

df_long = (
    df.select(["HeadIdx#"] + nozzle_cols)
    .melt(
        id_vars=["HeadIdx#"],
        value_vars=nozzle_cols,
        variable_name="nozzle_col",
        value_name="nozzle_value",
    )
    .with_columns(
        pl.col("nozzle_col")
          .str.extract(r"Value_(\d+)#_mean", group_index=1)
          .cast(pl.Int32)
          .alias("NozzleId")
    )
    .drop("nozzle_col")
    .rename({"HeadIdx#": "HeadIdx"})
    .drop_nulls()
)

df_agg = (
    df_long
    .group_by(["HeadIdx", "NozzleId"])
    .agg(pl.col("nozzle_value").std().alias("sd"))
```

```python
    .sort(["HeadIdx", "NozzleId"])
    .drop_nulls()
)

print(df_agg.head(10))
print(f"\nShape: {df_agg.shape}  |  HeadIdx range: {df_agg['HeadIdx'].min()}-
{df_agg['HeadIdx'].max()}")


x = df_agg["HeadIdx"].to_numpy()
y = df_agg["NozzleId"].to_numpy()
z = df_agg["sd"].to_numpy()


fig = plt.figure(figsize=(12, 8))
ax = fig.add_subplot(111, projection="3d")

sc = ax.scatter(x, y, z, c=z, cmap="plasma", s=18, alpha=0.85)

ax.set_xlabel("HeadIdx",  labelpad=10)
ax.set_ylabel("NozzleId", labelpad=10)
ax.set_zlabel("SD of nozzle value", labelpad=10)
ax.set_title("Nozzle SD by Head and Nozzle ID")

# Pivot to 2D grid for surface
pivot = df_agg.pivot(values="sd", index="HeadIdx", on="NozzleId").fill_null(0)
Z = pivot.select([c for c in pivot.columns if c != "HeadIdx"]).to_numpy()
X, Y = np.meshgrid(
    pivot["HeadIdx"].to_numpy(),
    np.arange(Z.shape[1])
)

ax.plot_surface(X, Y, Z.T, cmap="plasma", alpha=0.8)

plt.colorbar(sc, ax=ax, shrink=0.5, label="SD")
plt.tight_layout()
plt.show()
```